

3.3.1 Visual Studio Code (VS Code)

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 05 of 24

3.3.1 Visual Studio Code (VS Code)

LEARN IT

VS Code is a **free, lightweight and powerful code editor** made by Microsoft and widely used for Python. By default it is **not a full IDE**, but it becomes one when you install **extensions** (add-ons), such as the **Python extension by Microsoft**.

VS Code interface elements

Element	Description
1. Top Menu Bar	Control over files, editing, navigation and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains the core tools: Explorer, Search, Source Control, Run & Debug and Extensions.
3. Explorer	Shows project files and folders. Helps you create, open and manage your Python files.
4. Search	Finds and replaces text in the open file or across the whole project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes and collaborate.
6. Run & Debug	Runs code, finds errors, pauses at specific lines and lets you inspect variables.
7. Extensions	Like an App Store for VS Code. Add features such as the Python extension, Prettier or Python Debugger.



A simplified VS Code window: menu bar, side bar, Explorer and the code editor.

WORKED EXAMPLE

Setting up and running your first program in VS Code:

Install Python → Install VS Code → Extensions: install "Python" (Microsoft) → Explorer: new file hello.py → Type code → Click ▶ Run

```
print("Hello, World!")
```

OUTPUT
Hello, World!

The output appears in the **Terminal** panel at the bottom of the window.

REMEMBER IT

Extensions are the **"App Store"** of VS Code: just as apps give your phone new abilities, extensions give VS Code new abilities.

COMMON MISTAKE

Always save Python files with the `.py` extension (e.g. `hello.py`). Without it, VS Code does not know the file is Python, so there is no syntax highlighting and no Run button.

اردو میں سمجھیں

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔ یہ اپنے آپ میں مکمل آئی ڈی ای نہیں، لیکن ایکسٹینشنز (Extensions) لگانے سے آئی ڈی ای بن جاتا ہے۔ اس کے اہم حصے مینو بار، سائیڈ بار، ایکسپلورر، سرچ، سورس کنٹرول، رن اینڈ ڈیباگ اور ایکسٹینشنز ہیں۔

CHECK YOURSELF

1. Which VS Code element connects to Git?
2. Is VS Code a full IDE by default? How does it become one?
3. Where do you see and manage your project's files?

Answers: 1. Source Control. 2. No; it becomes an IDE by installing extensions (e.g. the Python extension). 3. Explorer.

Next lecture: 3.4 Fundamentals of Python Programming